



SMALLEST PORTABLE GAME BOY And Wi-Fi Hacking Device

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In the late 1980s and 1990s, the Game Boy redefined handheld entertainment. With its chunky buttons, monochrome screen, and pocket-sized design, it evolved from a toy into a cultural icon (see Fig. 1). For many, it was the first glimpse into portable electronics, sparking curiosity about gaming and the technology behind it. Decades later, the Game Boy continues to captivate retro gamers, engineers, and hobbyists, inspiring makers and hackers to reinvent its legacy.

Building on that legacy, the Smallest Portable Game Boy Retro Console and Wi-Fi Hacking Device (see Fig. 1) unites retro gaming charm with modern embedded technology. Powered by the ESP12F Wi-Fi module (ESP8266), it enables

retro-style gaming, Wi-Fi scanning, deauthentication, and Wi-Fi penetration testing. Its compact 4.5cm round PCB packs all essentials—an LDO regulator, capacitors, resistors, a battery connector with power switch, and tactile control buttons.

Display options include an OLED SSD1306 for a classic 90s look or a TFT LCD for modern visuals. Supported by open source libraries, the device allows development of custom games or deployment of prebuilt hacking tools, making it a versatile, retro-inspired, future-ready device for makers and security researchers.

The ESP12F chip delivers the processing power required for running Game Boy games while supporting Wi-Fi hacking, making

it ideal for dual-use designs. It provides sufficient I/O, memory, and a compact footprint that aligns with system requirements.



Fig. 1: Author's prototype



Fig. 2: Game Boy available in the market vs the author's Game Boy design

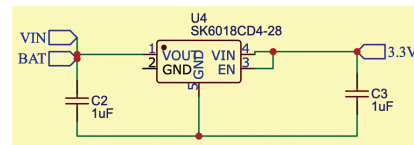


Fig. 3: LDO voltage regulator circuit

TABLE 1
BILL OF MATERIALS

Name	Designator	Footprint	Quantity	Manufacturer Part	Manufacturer
6X6-PUSH BUTTON COPY	BUT1, BUT2, BUT3, BUT4	6X6-PUSH BUTTON	4		
100nF	C1	C0402	1	GRM-155R71H104KE14D	muRata
1μF	C2, C3	C0402	2	CL05A105KP5NNNC	SAMSUNG
SM02B-SRSS-TB (LF)(SN)	CN1	CONN-SMD_2P-P1.00_SM02B-SRSS-TB-LF-SN	1	SM02B-SRSS-TB (LF)(SN)	JST
OLED DISPLAY	LCD1	SSD1306-OLED-128X64-I2C	1		
ESP-12F	MOD1	WIFIM-SMD ESP-12F-ESP-8266MOD	1	ESP-12F (ESP-8266MOD)	Ai-Thinker
10kΩ	R1, R2	R0603	2	RC0603FR-0710KL	YAGEO
MK-11C04-G013	SW2	SW-SMD_MK-11C04-G013	1	MK-11C04-G013	G-Switch
St7789v	U1	1.28-inch Round Edge Display	1	19192	Waveshare
USB 2.0 Solder Pad	U3	USB SOLDER PADS	1		
SK6018CD4-28	U4	DFN-4_L1.0-W1.0-P0.65-BL-EP	1	SK6018CD4-28	suntek